Practical Digital Systems Design

Homework 1:

Gate-Level Design & Analysis of a

14-bit Combinational Adder/Subtractor

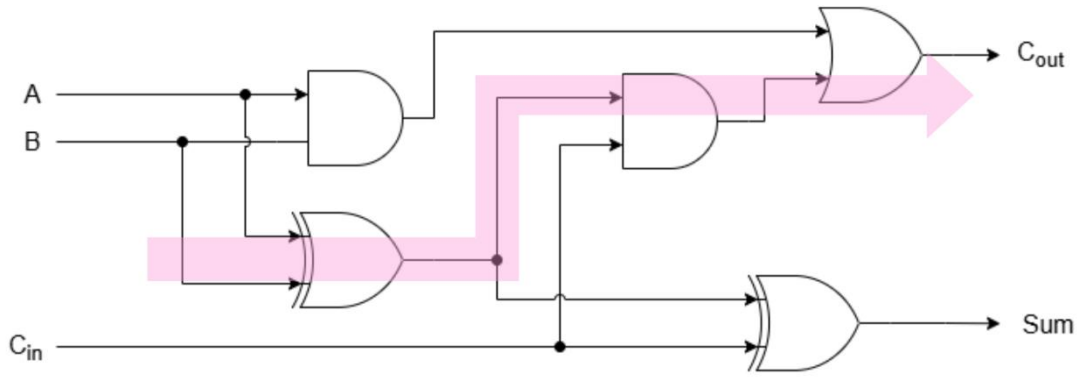
Name: 尋以諾

Student ID: B123015021

I. Design Principle

The purpose of the assignment is to design a 14-bit combinational adder/subtractor using gate-level Verilog description.
The circuit performs addition or subtraction depending on the control signal “mode”, where mode = 1 selects addition and mode = 0 selects subtraction. The subtraction is implemented through the two’s-complement method, in which input B is inverted and one is added through the carry-in.

1. Ripple-Carry Adder (RCA)
   A. Full Adder (FA)
      The full adder is the fundamental building block of the ripple-carry adder.



It performs a 1-bit addition of two operands (A, B) and a carry-in (C_{in}), producing a sum output (Sum) and a carry-out (C_{out}).

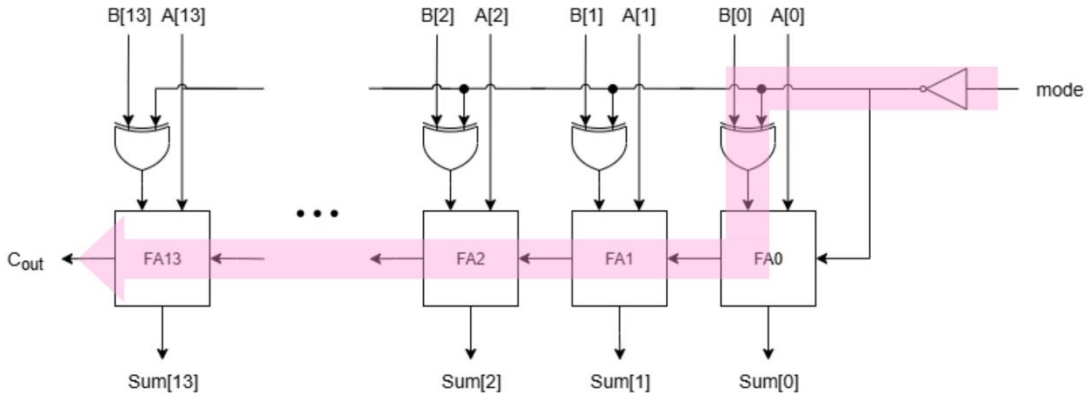
The Boolean expressions of a full adder:

$$S = A \oplus B \oplus C_{in}$$

$$C_{out} = A \cdot B + C_{in} \cdot (A \oplus B)$$

i. Logic gates: 5
ii. Logic levels: 2
iii. Critical path: 3 gates (as indicated by the pink arrow)

B. 14-bit Adder/Subtractor



A 14-bit ripple-carry adder is constructed by cascading fourteen full adders in sequence, where the carry-out of each stage serves as the carry-in of the next stage.

To enable subtraction, each bit of input B is XORed with the complement of the control signal (~mode), and the initial carry-in is also set to it.

$$B' = B \oplus (\sim mode), \qquad Cin = \sim mode$$

$$Sum = A + B' + Cin$$

i. Logic gates: $14 \times 1 + 14 \times 5 + 1 = 85$

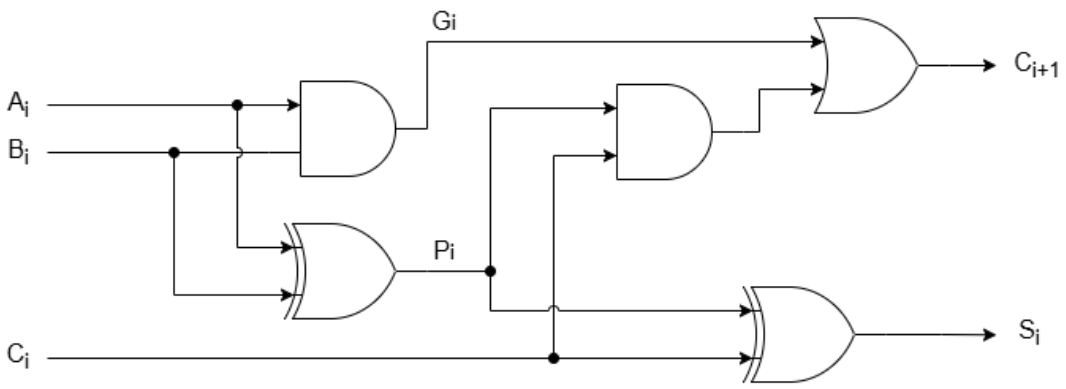
ii. Logic levels: $1 + 14 \times 2 = 29$

iii. Critical Path: $1 + 1 + 3 + 2 \times 13 = 31$ gates
(as indicated by the pink arrow)

C. Summary

The ripple-carry adder has the advantages of simple structure, low hardware cost, and easy implementation, making it ideal for small-scale arithmetic circuits. However, because each carry signal must propagate sequentially through all stages, the overall delay increases linearly with bit width, resulting in poor speed performance for larger designs.

2. Carry-Lookahead Adder (CLA)
   A. Propagate (P) & Generate (G)
      The CLA computes carry bits in parallel by generating propagate and generate signals for each bit:



$$P_i = A_i \oplus B_i, \qquad G_i = A_i \cdot B_i$$

$$C_{i+1} = G_i + P_i \cdot C_i$$

By expanding the recursive carry relation, all carries can be determined within a few logic levels, reducing the overall delay to a near-constant value regardless of bit-width.

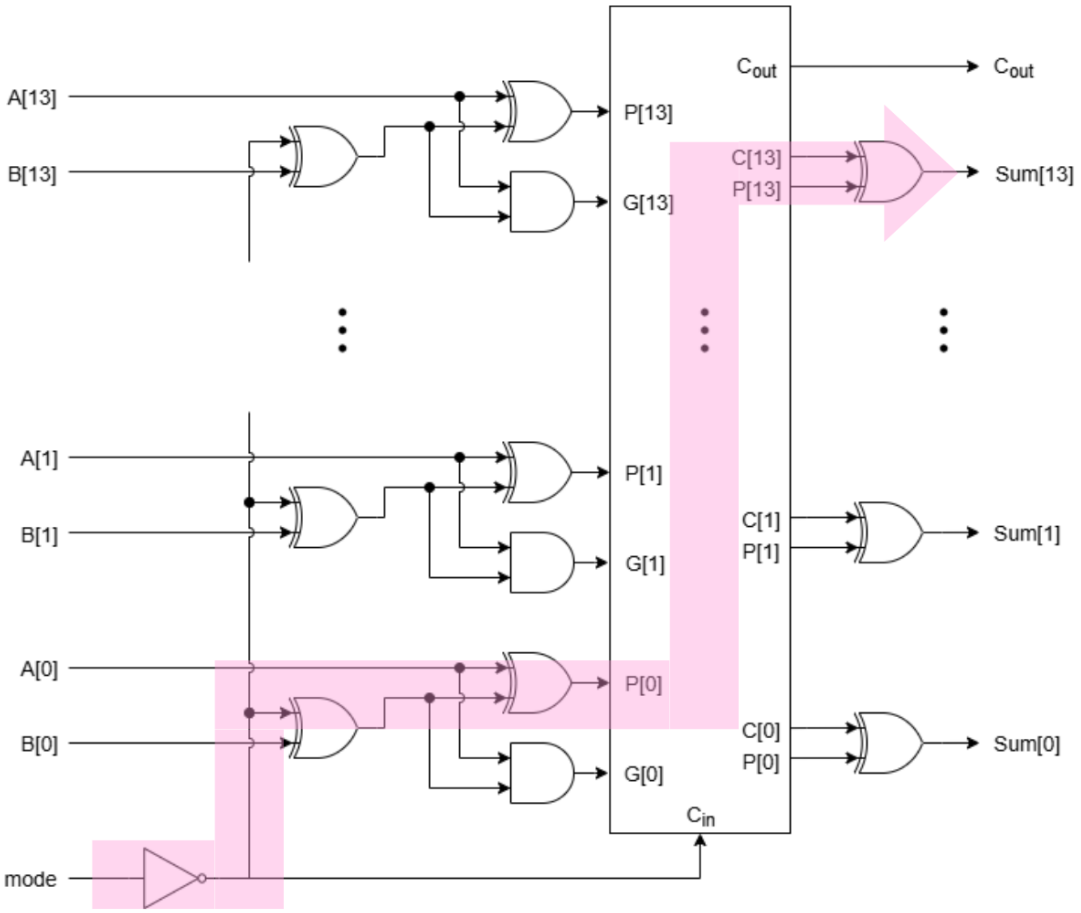
$$C_1 = G_0 + P_0 C_0$$

$$C_2 = G_1 + P_1 C_1 = G_1 + P_1 G_0 + P_1 P_0 C_0$$

$$C_3 = G_2 + P_2 C_2 = G_2 + P_2 G_1 + P_2 P_1 G_0 + P_2 P_1 P_0 C_0$$

$$\vdots$$

   B. 14-bit Adder/Subtractor



For a 14-bit adder constructed from smaller CLA blocks (for example, three 4-bit CLAs and one 2-bit CLA), the total gate count can be estimated as follows:

i. Logic gates: $1 + 14 \times 3 + 3 \times 14 + 1 \times 5 + 14 = 104$
ii. Logic levels: 8
iii. Critical path: 9 gates (as indicated by the pink arrow)

C. Summary

The carry-lookahead adder achieves high-speed operation by generating carry signals in parallel using propagate and generate logic, rather than allowing them to ripple through each stage sequentially.

However, it requires a large number of gates and additional wiring to form the carry-lookahead network, leading to a substantially higher gate count and larger circuit area.

3. Overflow Control

   For signed arithmetic, overflow occurs when the carry into the most significant bit differs from the carry out of that bit.

   This can be easily detected using a single XOR gate:

$$Overflow = C_{13} \oplus C_{out}$$

II. Result Analysis

1. Unsigned

   A. Addition (mode = 1)

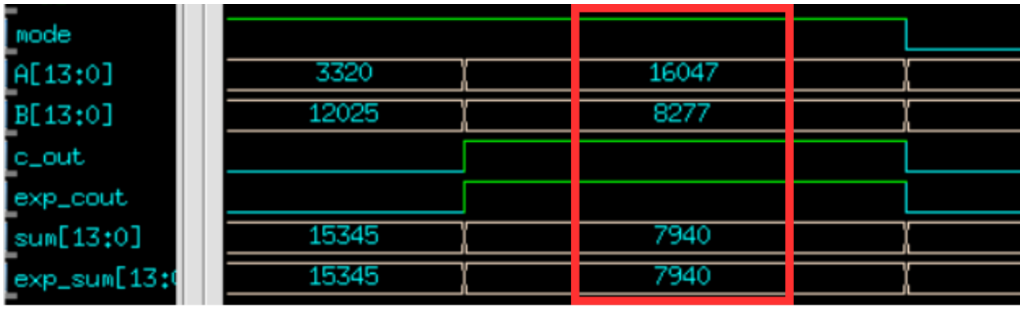
      i. Positive + Positive (No Overflow)



$$2254 + 45 = 2299$$

$$\rightarrow sum = 2299, \quad c_{out} = 0$$

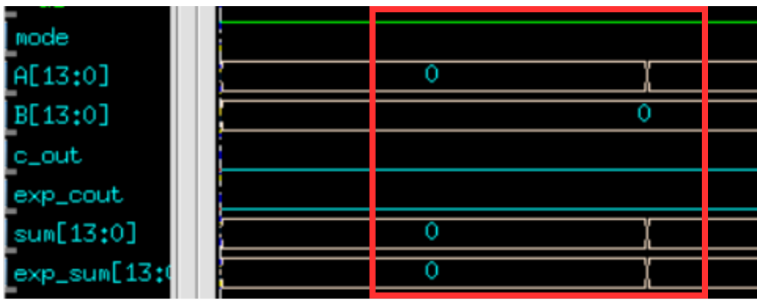
      ii. Positive + Positive (Overflow)



$$16047 + 8277 = 24324 \rightarrow 7940$$

$$\rightarrow sum = 7940, \quad c_{out} = 1$$

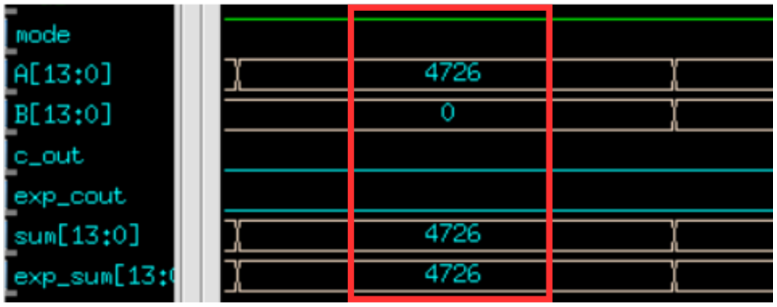
iii. Zero + Zero



$$0 + 0 = 0$$

$$\rightarrow sum = 0, \quad c_{out} = 0$$

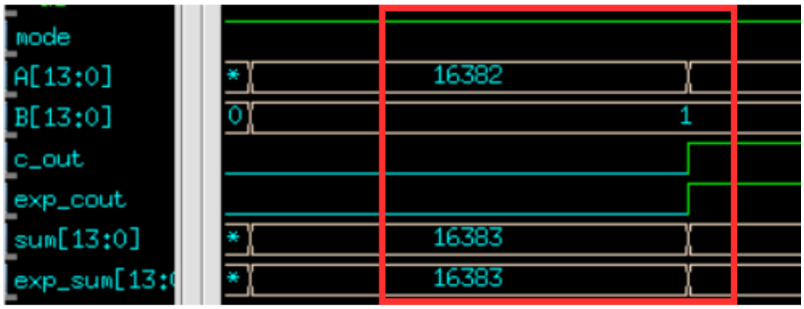
iv. Identity



$$4726 + 0 = 4726$$

$$\rightarrow sum = 4726, \quad c_{out} = 0$$

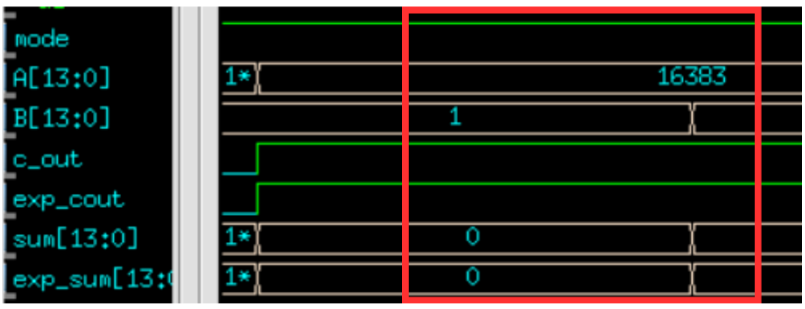
v. Boundary Limit (No Overflow)



$$16382 + 1 = 16383 < 16384$$

$$\rightarrow sum = 16383, \quad c_{out} = 0$$

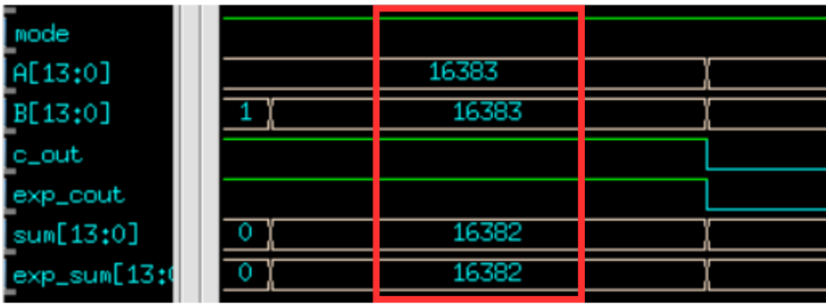
vi. Boundary Limit (Overflow)



$$16383 + 1 = 16384 \geq 16384 \rightarrow 0$$

$$\rightarrow sum = 0, \quad c_{out} = 1$$

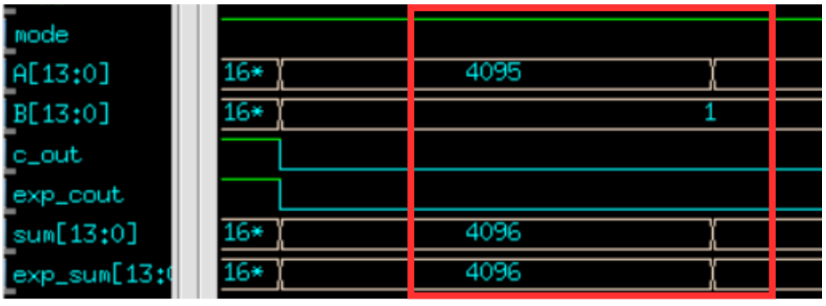
vii. Extreme Overflow



$$16383 + 16383 = 32766 \geq 16384 \rightarrow 16382$$

$$\rightarrow sum = 16382, \quad c_{out} = 1$$

viii. Long Carry Chain

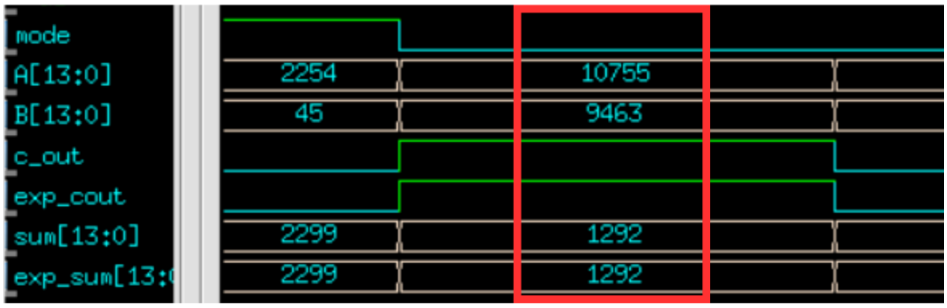


$$4095 + 1 = 4096$$

$$\rightarrow sum = 4096, \quad c_{out} = 0$$

B. Subtraction (mode = 0)

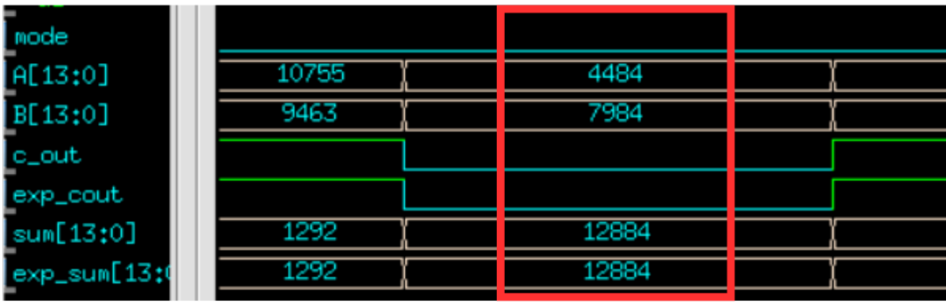
i. Positive – Positive (Positive Result)



$$10755 - 9463 = 1292$$

$$\rightarrow sum = 1292, \quad c_{out} = 1$$

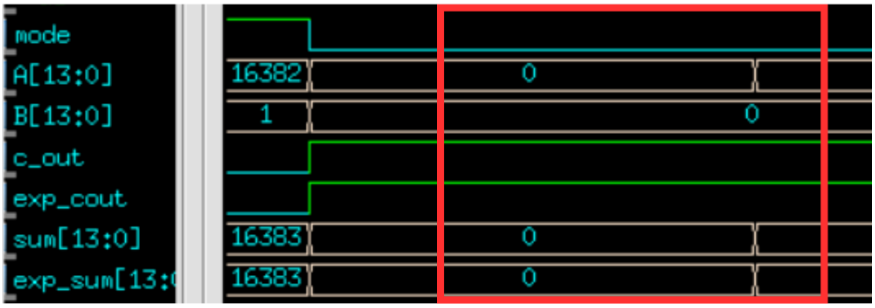
ii. Positive – Positive (Negative Result)



$$4484 - 7984 = -3500 \rightarrow 12884 \ \ (2's \ \ complement)$$

$$\rightarrow sum = 12884, \quad c_{out} = 0$$

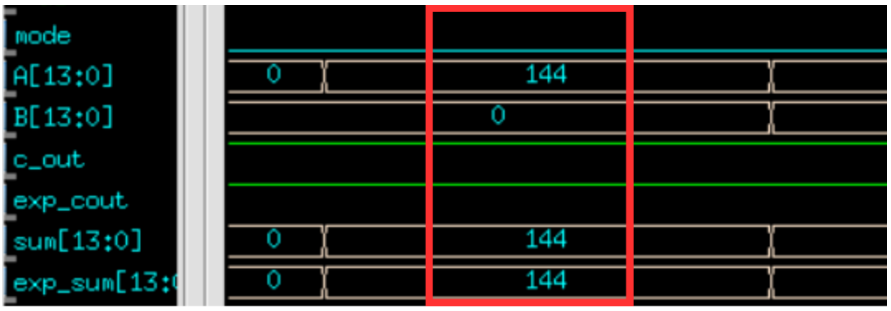
iii. Zero – Zero



$$0 - 0 = 0$$

$$\rightarrow sum = 0, \quad c_{out} = 1$$

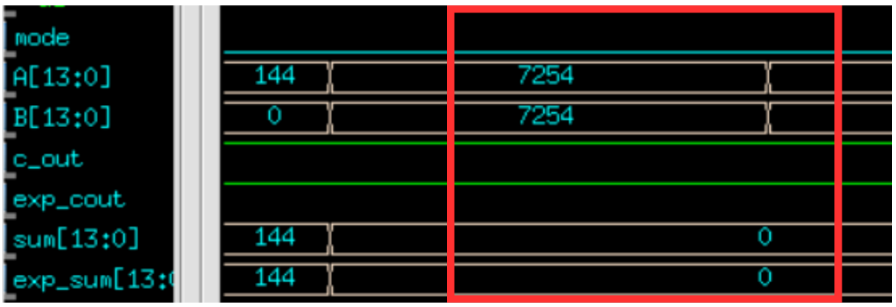
iv. Identity



$$144 - 0 = 144$$

$$\rightarrow sum = 144, \quad c_{out} = 1$$

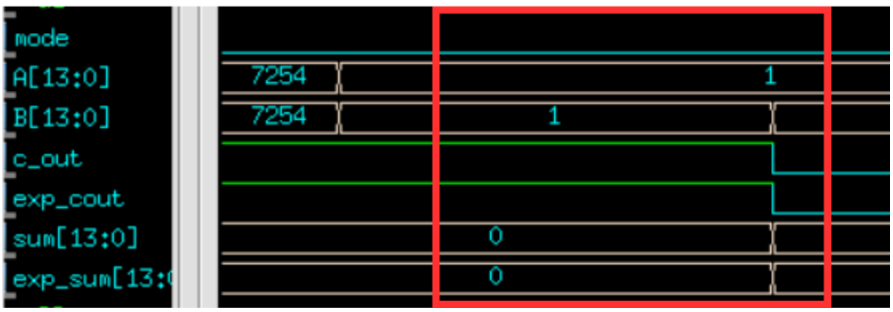
v. X – X



$$7254 - 7254 = 0$$

$$\rightarrow sum = 0, \quad c_{out} = 1$$

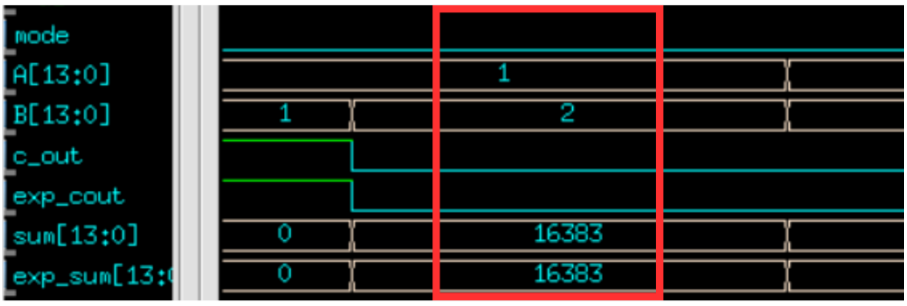
vi. Boundary Limit (No Overflow)



$$1 - 1 = 0$$

$$\rightarrow sum = 0, \quad c_{out} = 1$$

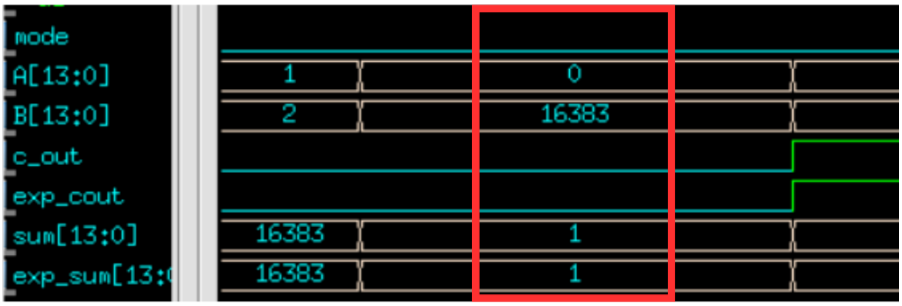
vii. Boundary Limit (Overflow)



$$1 - 2 = -1 \rightarrow 16383 \ \ (2's \ \ complement)$$

$$\rightarrow sum = 16383, \quad c_{out} = 0$$

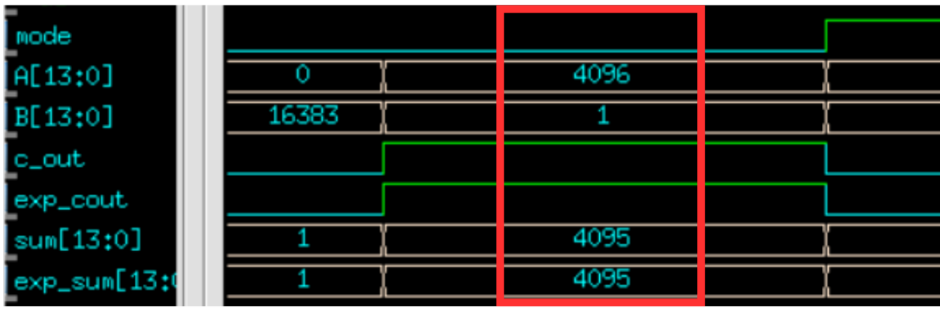
viii. Extreme Overflow



$$0 - 16383 = -16383 \rightarrow 1 \ \ (2's \ \ complement)$$

$$\rightarrow sum = 1, \quad c_{out} = 0$$

ix. Long Borrow Chain



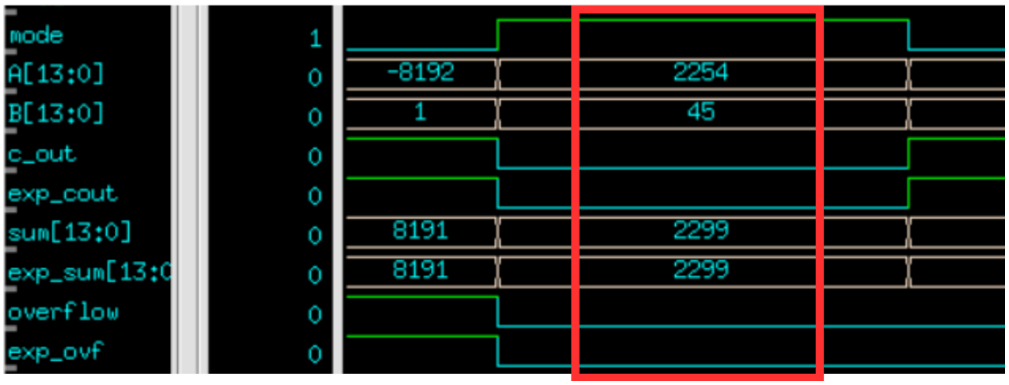
$$4096 - 1 = 4095$$

$$\rightarrow sum = 4095, \quad c_{out} = 1$$

2. Signed

A. Addition (mode = 1)

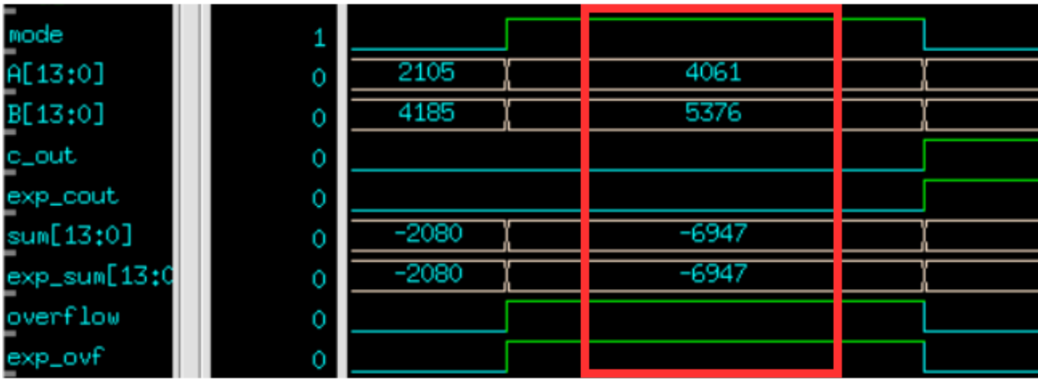
i. Positive + Positive (No Overflow)



$$2254 + 45 = 2299$$

$$\rightarrow sum = 2299, \quad c_{out} = 0, \quad overflow = 0$$

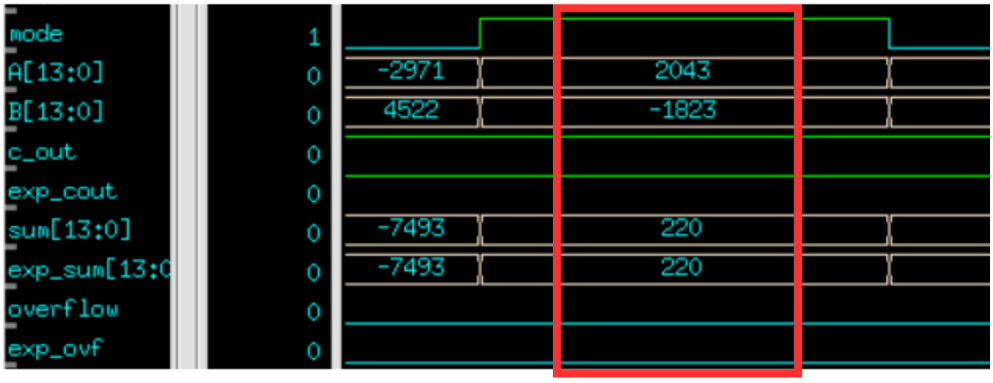
ii. Positive + Positive (Overflow)



$$4061 + 5376 = 9437 \geq 8192 \rightarrow -6947$$

$$\rightarrow sum = -6947, \quad c_{out} = 0, \quad overflow = 1$$

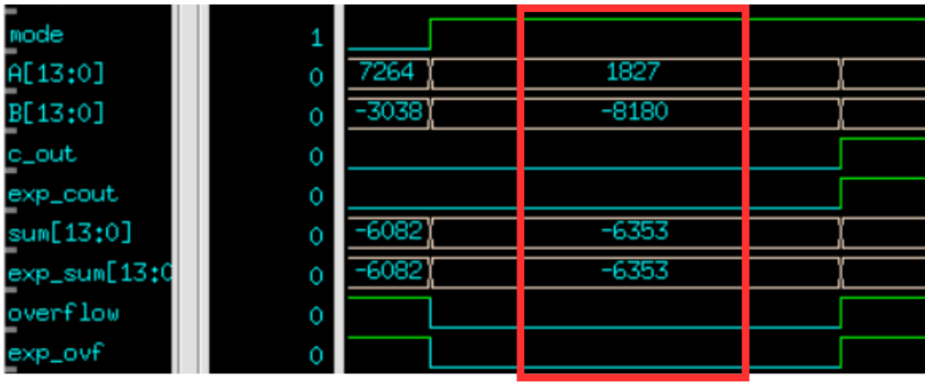
iii. Positive + Negative (Positive Result)



$$2043 + (-1823) = 220$$

$$\rightarrow sum = 220, \quad c_{out} = 1, \quad overflow = 0$$

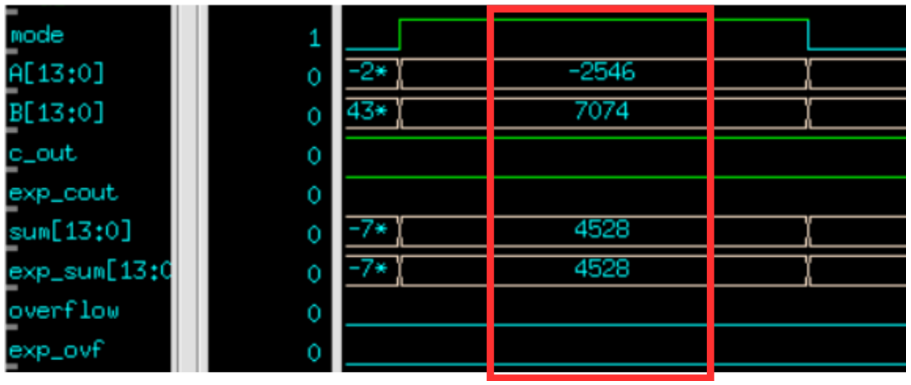
iv. Positive + Negative (Negative Result)



$$1827 + (-8189) = -6353$$

$$\rightarrow sum = -6353, \quad c_{out} = 0, \quad overflow = 0$$

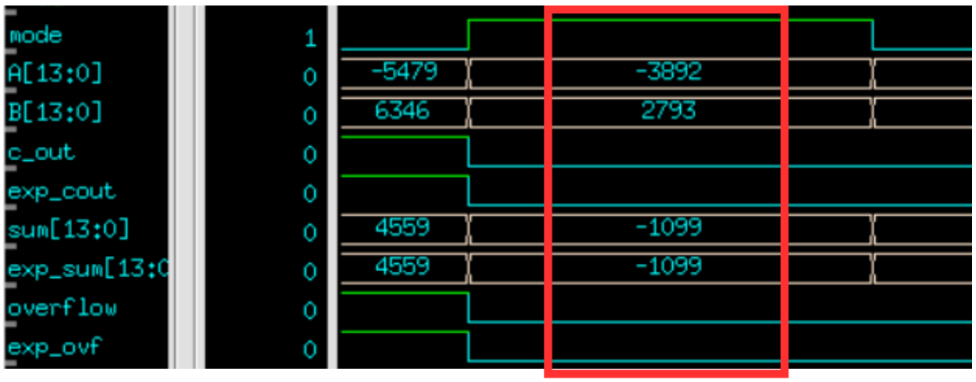
v. Negative + Positive (Positive Result)



$$-2546 + 7074 = 4528$$

$$\rightarrow sum = 4528, \quad c_{out} = 1, \quad overflow = 0$$

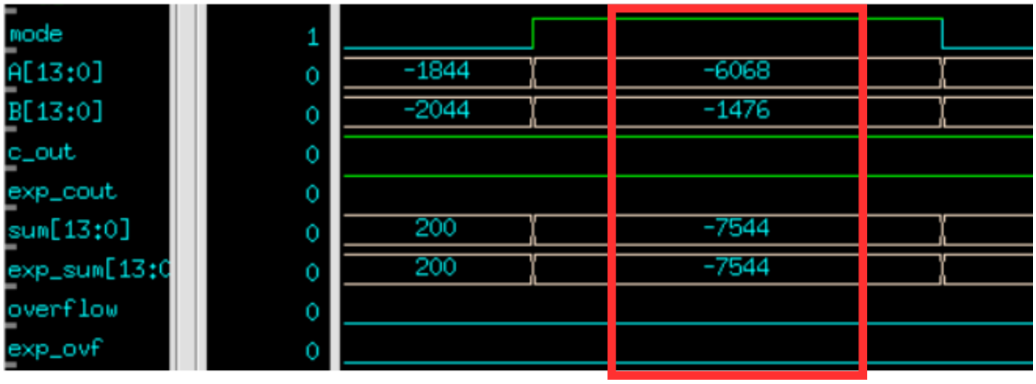
vi. Negative + Positive (Negative Result)



$$-3892 + 2793 = -1099$$

$$\rightarrow sum = -1099, \quad c_{out} = 0, \quad overflow = 0$$

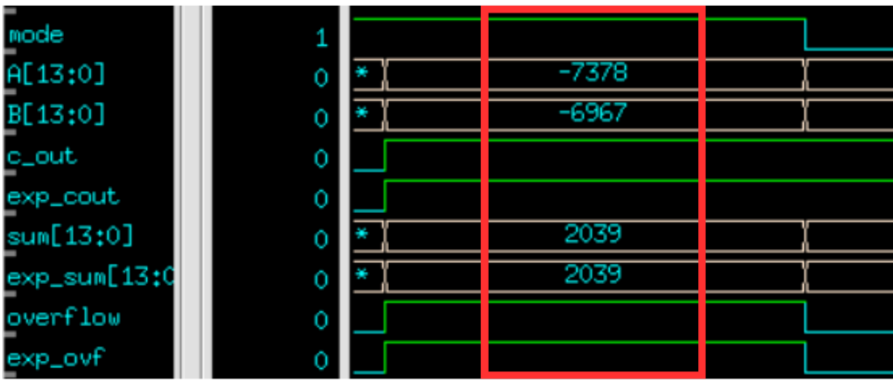
vii. Negative + Negative (No Overflow)



$$-6068 + (-1476) = -7544$$

$$\rightarrow sum = -7544, \quad c_{out} = 1, \quad overflow = 0$$

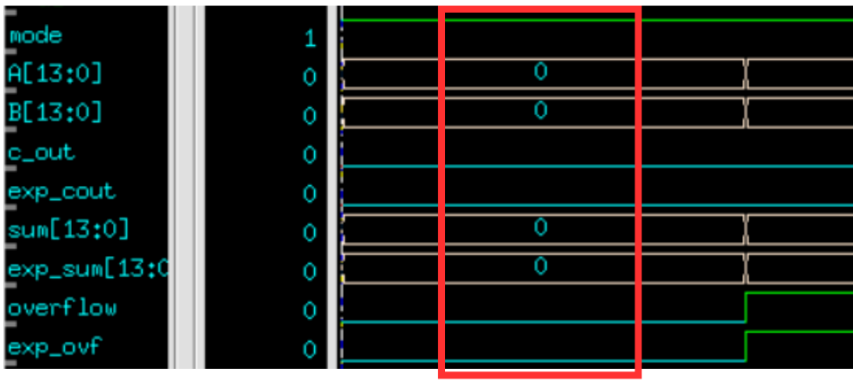
viii. Negative + Negative (Overflow)



$$-7378 + (-6967) = -14345 \leq -8193 \rightarrow 2039$$

$$\rightarrow sum = 2039, \quad c_{out} = 1, \quad overflow = 1$$

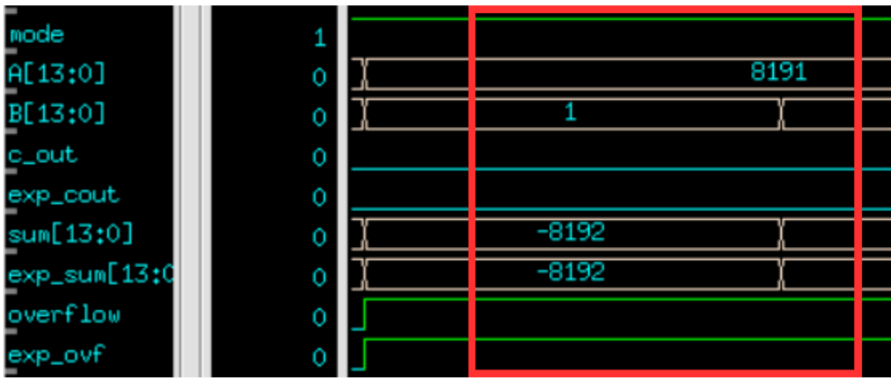
ix. Zero + Zero



$$0 + 0 = 0$$

$$\rightarrow sum = 0, \quad c_{out} = 0, \quad overflow = 0$$

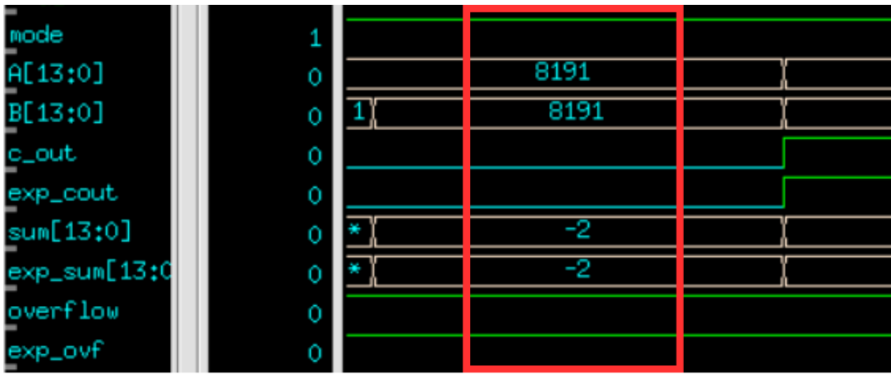
x. Upper Bound Limit (Overflow)



$$8191 + 1 = 8192 \geq 8192 \rightarrow -8192$$

$$\rightarrow sum = -8192, \quad c_{out} = 0, \quad overflow = 1$$

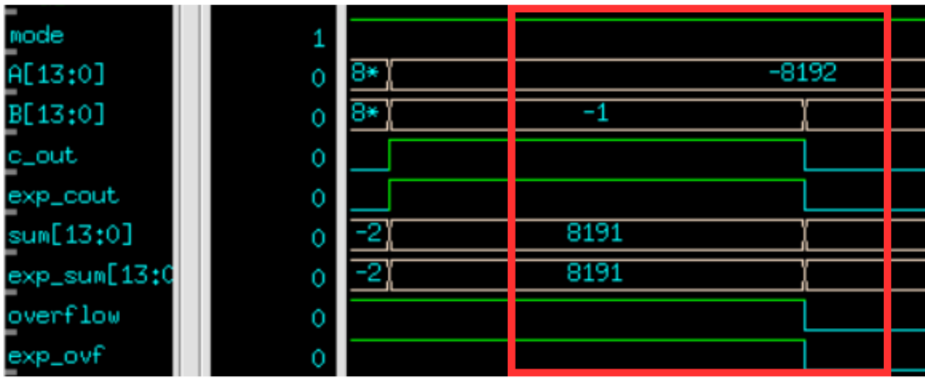
xi. Extreme Overflow



$$8191 + 8191 = 16382 \geq 8192 \rightarrow -2$$

$$\rightarrow sum = -2, \quad c_{out} = 0, \quad overflow = 1$$

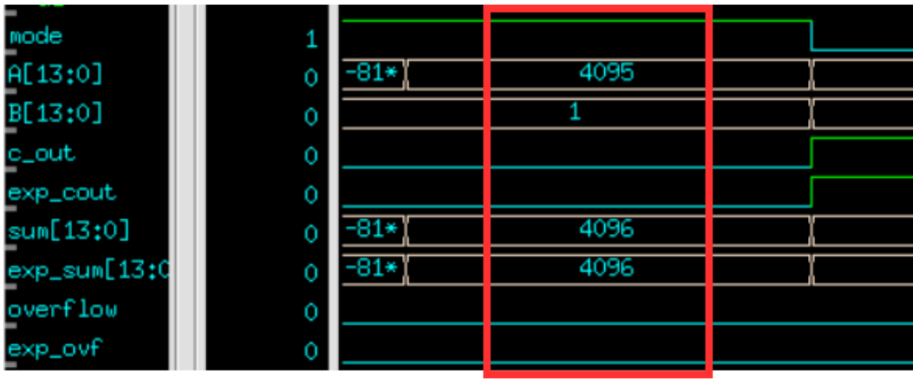
xii. Lower Bound Limit (Overflow)



$$-8192 + (-1) = -8193 \leq -8193 \rightarrow 8191$$

$$\rightarrow sum = 8191, \quad c_{out} = 1, \quad overflow = 1$$

xiii. Long Carry Chain

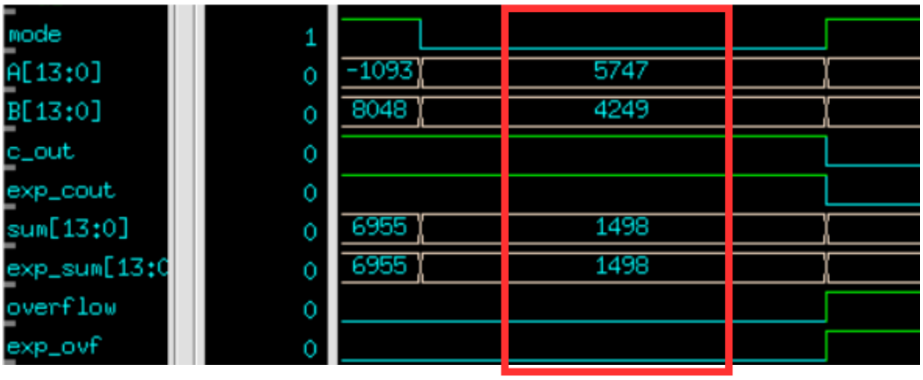


$$4095 + 1 = 4096$$

$$\rightarrow sum = 4096, \quad c_{out} = 0, \quad overflow = 0$$

B. Subtraction (mode = 0)

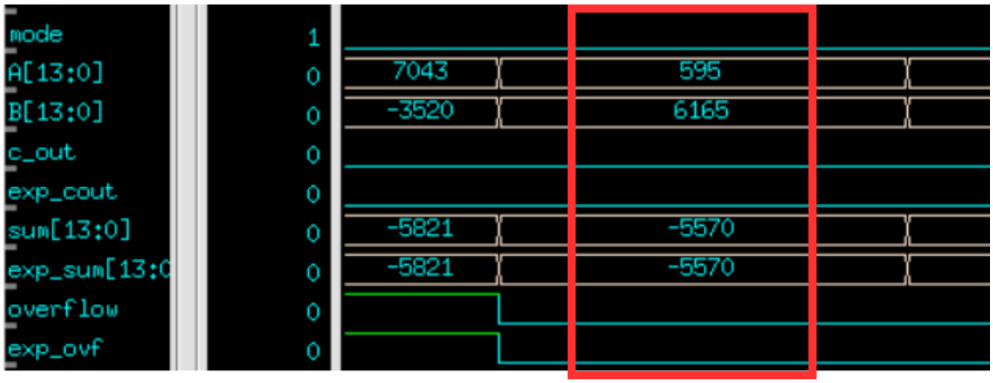
i. Positive – Positive (Positive Result)



$$5747 - 4249 = 1498$$

$$\rightarrow sum = 1498, \quad c_{out} = 1, \quad overflow = 0$$

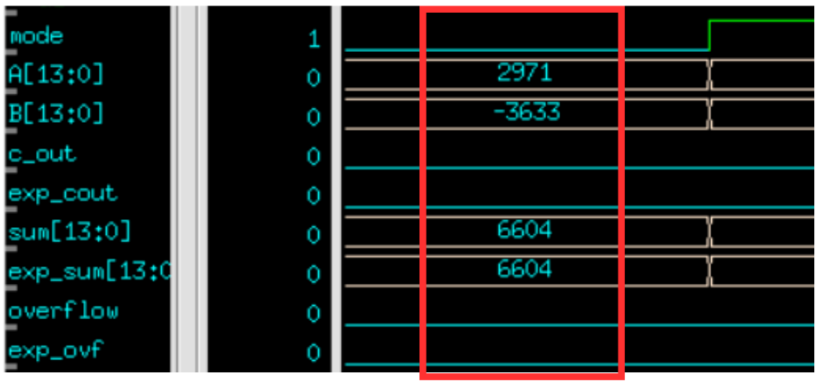
ii. Positive – Positive (Negative Result)



$$595 - 6165 = -5570$$

$$\rightarrow sum = -5570, \quad c_{out} = 0, \quad overflow = 0$$

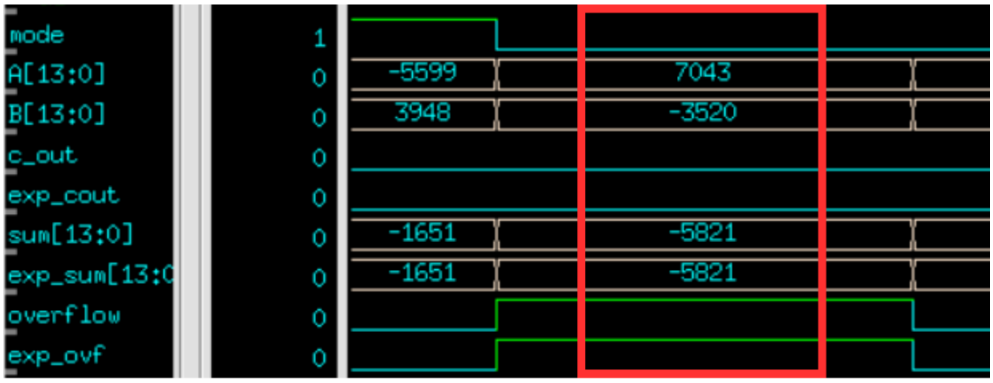
iii. Positive – Negative (No Overflow)



$$2971 - (-3633) = 6604$$

$$\rightarrow sum = 6604, \quad c_{out} = 0, \quad overflow = 0$$

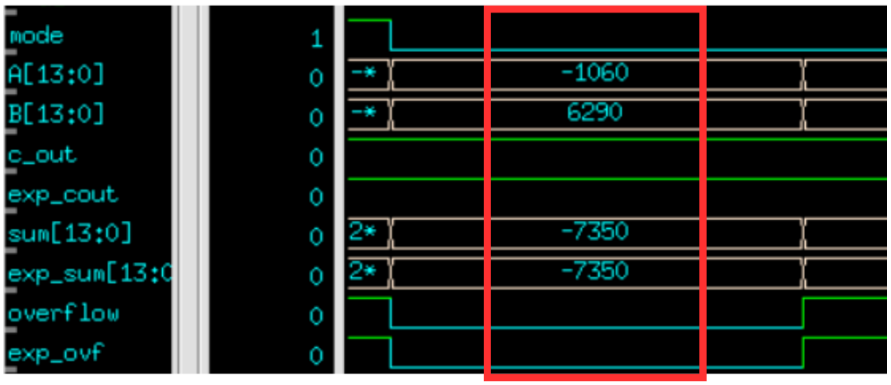
iv. Positive – Negative (Overflow)



$$7043 - (-3520) = 10563 \geq 8192 \rightarrow -5821$$

$$\rightarrow sum = -5821, \quad c_{out} = 0, \quad overflow = 1$$

v. Negative – Positive (No Overflow)



$$-1060 - 6290 = -7350$$

$$\rightarrow sum = -7350, \quad c_{out} = 1, \quad overflow = 0$$

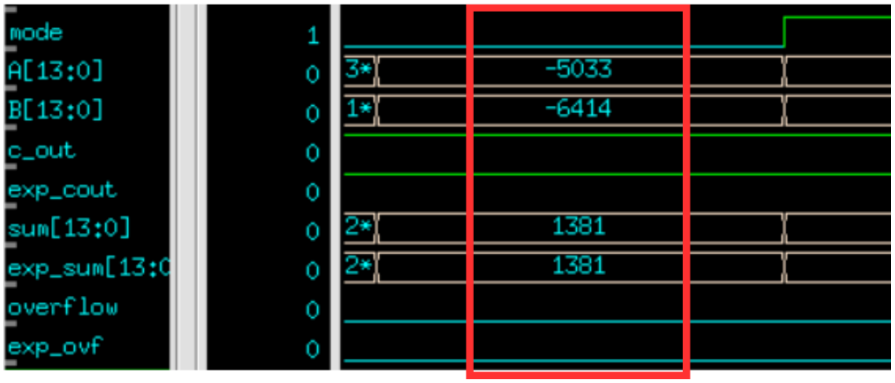
vi. Negative – Positive (Overflow)



$$-4510 - 4872 = -9382 \leq -8193 \rightarrow 7002$$

$$\rightarrow sum = 7002, \quad c_{out} = 1, \quad overflow = 1$$

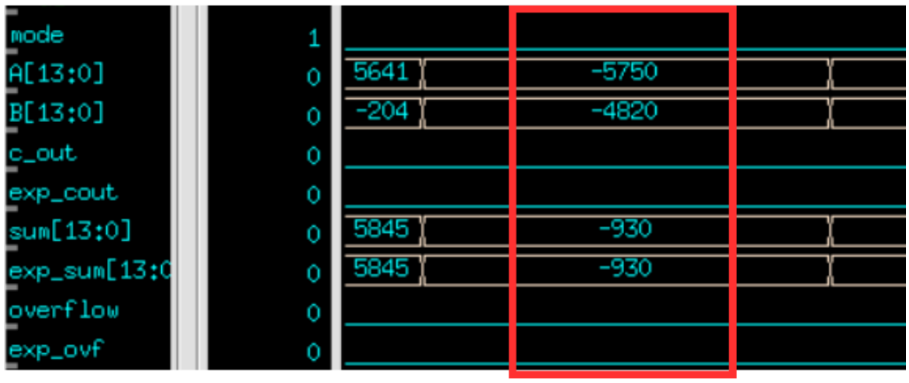
vii. Negative – Negative (Positive Result)



$$-5033 - (-6414) = 1381$$

$$\rightarrow sum = 1381, \quad c_{out} = 1, \quad overflow = 0$$

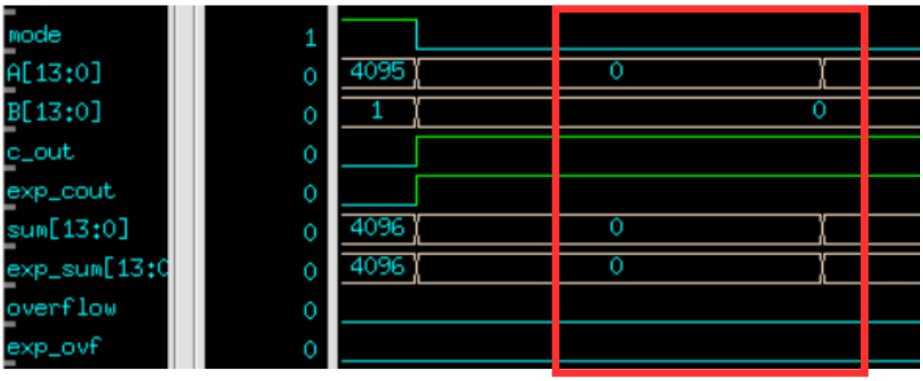
viii. Negative – Negative (Overflow)



$$-5750 - (-4820) = -930$$

$$\rightarrow sum = -930, \quad c_{out} = 0, \quad overflow = 0$$

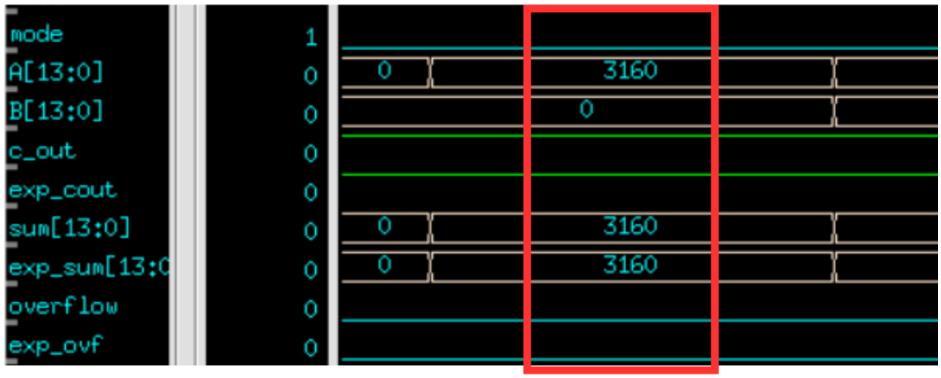
ix. Zero – Zero



$$0 - 0 = 0$$

$$\rightarrow sum = 0, \quad c_{out} = 1, \quad overflow = 0$$

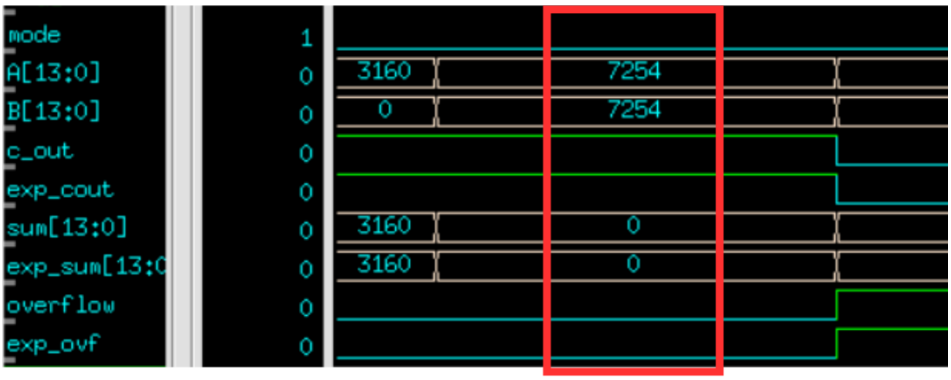
x. Identity



$$3160 - 0 = 3160$$

$$\rightarrow sum = 3160, \quad c_{out} = 1, \quad overflow = 0$$

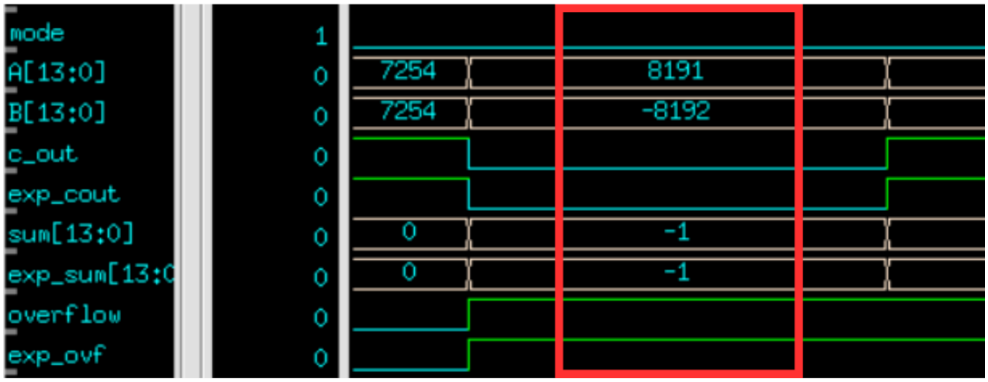
xi. X – X



$$7254 - 7254 = 0$$

$$\rightarrow sum = 0, \quad c_{out} = 1, \quad overflow = 0$$

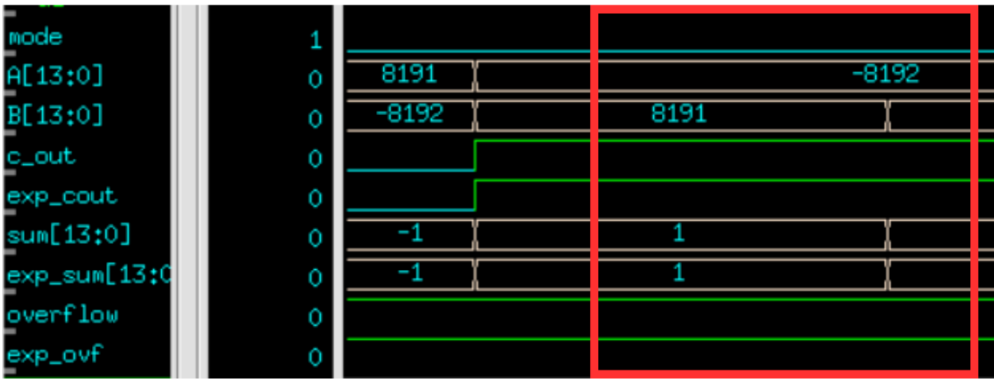
xii. Extreme Positive Overflow



$$8191 - (-8192) = 16383 \geq 8192 \rightarrow -1$$

$$\rightarrow sum = -1, \quad c_{out} = 0, \quad overflow = 1$$

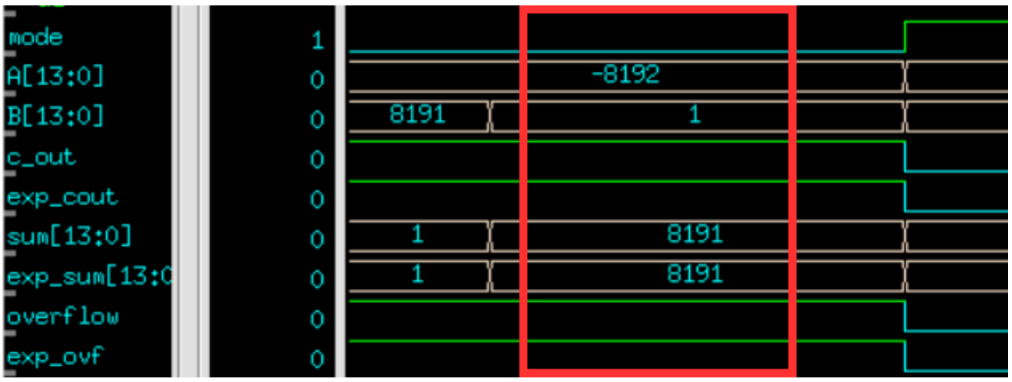
xiii. Extreme Negative Overflow



$$-8192 - 8191 = -16383 \leq -8193 \rightarrow 1$$

$$\rightarrow sum = 1, \quad c_{out} = 1, \quad overflow = 1$$

xiv. Lower Bound Limit (Overflow)



$$-8192 - 1 = -8193 \leq -8193 \rightarrow 8191$$

$$\rightarrow sum = 8191, \quad c_{out} = 1, \quad overflow = 1$$

III. Testbench Design

The testbench was designed to verify the functionality of the 14-bit adder/subtractor using both directed and random test cases under unsigned and signed modes. Directed tests focused on corner cases such as zero operations, overflow and borrow boundaries, maximum and minimum signed values, and long carry/borrow chains, while 500 random vectors ensured broad coverage. The testbench included an automatic checking mechanism that computed the expected result, compared it to the circuit output, and counted passes and fails. Whenever a mismatch occurred, the testbench displayed detailed information, including the operation type, input values, expected result, and circuit output for debugging purposes. After all tests completed successfully with zero failures, the simulation printed a clear “ALL PASS” message along with a visual banner.



IV. Discussion & Reflection

One interesting challenge I encountered was when I initially placed the delay in front of the expected value calculation in my testbench. This caused the reference results to always appear one time unit after the circuit output in the waveform, making it seem like the adder was producing incorrect results even though the logic was correct.

Another major takeaway came from writing the gate-level code of the carry-lookahead adder. While its advantage in speed is clear compared to the ripple-carry adder, I observed how rapidly the complexity of the logic grows with each additional bit. The number of generate and propagate signals, as well as the internal wires, grows exponentially, making the schematic difficult to trace.

Overall, this homework helped me become more familiar with using NCVerilog for verification and nWave for waveform analysis. I also gained a clearer understanding of different adder architectures, showing the trade-off between circuit simplicity and computation speed.